

MANAS RANJAN BARIK

Software Engineer

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SUMMARY

Software Engineer with 1.7+ years of experience developing backend services and full-stack applications for a multi-tenant SaaS platform. Proficient in TypeScript, JavaScript, Node.js, NestJS, React, and PostgreSQL, with hands-on experience building microservices, REST APIs, authentication systems, access control frameworks, and data integrations. Skilled at designing secure, scalable, and maintainable software solutions that improve platform reliability, operational efficiency, and user experience.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, SQL
- **Backend:** Node.js, Express, NestJS, REST APIs, Microservices
- **Frontend:** React, Vite, Micro-frontends (Module Federation)
- **Databases:** PostgreSQL, Prisma (ORM), Redis
- **Auth & Security:** Role-Based Access Control (RBAC), Row-Level Security (RLS), JWT
- **Data & Integrations:** Snowflake, Excel / data ingestion, DuckDB, Parquet
- **ORM & Tooling:** TypeORM, Prisma, Git
- **Practices:** Multi-tenant architecture, API design

EXPERIENCE

DataCaffe | Software Engineer | Dec 2024 - Present

- Built and enhanced backend services for a multi-tenant BI platform using Node.js, TypeScript, PostgreSQL, and microservices architecture.
- Implemented RBAC, row-level security, JWT authentication, and workspace isolation mechanisms, strengthening platform security and protecting tenant data.
- Developed Excel and Snowflake integrations with schema synchronization and retry logic, reducing failed imports and improving data ingestion reliability for customer workspaces.
- Reduced failed data imports by 40% through automated retry logic and schema synchronization for Snowflake and Excel connectors.
- Decreased administrative effort by 50% by building a dedicated admin console for user, role, and platform configuration management.
- Secured access for 100+ users across multiple workspaces by implementing RBAC and row-level security controls.
- Developed per-user API and LLM token usage tracking and quota management, improving cost visibility and enabling controlled resource consumption across customer accounts.

PROJECTS

RAG Tutor & Plagiarism Detector | Python, FastAPI, LangChain, FAISS, Ollama, Streamlit

- Reduced LLM hallucinations in a Q&A tutor by building a RAG pipeline (FAISS + LangChain) with a constrained prompt that answers only from retrieved text and flags when no answer exists.
- Improved plagiarism detection accuracy by combining three checks — TF-IDF for direct copying, embeddings for paraphrased text, and an LLM-based AI-text check — into one weighted 0–100 integrity score with clean / review / flagged verdicts.
- Removed API cost and kept student data local by running everything on open models (Ollama), with graceful fallbacks so the FastAPI / Streamlit services stay usable when the model backend is down.

EDUCATION

GITA Autonomous College — B.Tech, Electrical Engineering

Bhubaneswar, Odisha | CGPA: 7.71

2019 – 2023